

Diffusion-based Image Super-Resolution: A Comparative Study Against GANs

Shaharyar Rizwan, Ahmed Ali Zahid, and Naveed Ahmed Bajwa

Department of Computer Science, National University of Computer and Emerging Sciences

{i220999, i220889, i221271}@nu.edu.pk

Abstract. Single Image Super-Resolution (SISR) is one of the main core tasks in computer vision which is aimed at recovering high-resolution images from low-resolution observations. Although Generative Adversarial Networks (GANs) have for the longest time been the state-of-the-art go to models for generating perceptually pleasing results, Denoising Diffusion Probabilistic Models (DDPMs) have recently emerged as a strong and worthy alternative. This paper presents a comparative study evaluating the performance, perceptual quality, and computational efficiency of diffusion-based models (DDPM and DDIM) against a strong GAN baseline (ESRGAN). We have implemented a U-Net-based diffusion model which has been conditioned on bicubic upsampling and we have compared it against a pretrained ESRGAN model fine-tuned on the 800-image DIV2K corpus. Our experiments demonstrate that while ESRGAN achieves superior Peak Signal-to-Noise Ratio (PSNR) and Learned Perceptual Image Patch Similarity (LPIPS) scores (27.76 dB and 0.22 on Set5, respectively) with real-time inference speeds (0.06s/image), diffusion models offer an inherently different stochastic refinement pathway. We also further conduct an ablation study on the diffusion sampler steps and noise schedules, in order to study and highlight the inherent trade-offs between sampling speed and reconstruction fidelity, and confirm that DDIM cuts latency by roughly $20\times$ relative to full-chain DDPM sampling.

Keywords: Generative AI Image Super-Resolution Diffusion Models
ESRGAN Deep Learning.

1 Introduction

Single Image Super-Resolution(SISR) which is the recovery of high-resolution (HR) images from low-resolution (LR) inputs, is an ill-posed inverse problem. Traditional interpolation methods, such as bicubic upsampling, very often lead to blurry and overly smoothed images. Deep learning approaches, especially Convolutional Neural Networks (CNNs) optimized for mean squared error (MSE), improve fidelity but they also fall short and struggle to generate high-frequency textures. To address this issue, Generative Adversarial Networks (GANs) like SRGAN and ESRGAN were introduced, they helped to dramatically improve

perceptual quality by optimizing an adversarial loss rather than purely pixel-wise differences [1].

More recently, Denoising Diffusion Probabilistic Models (DDPMs) [2] have revitalized the field of generative AI, demonstrating unprecedented image synthesis capabilities. Models such as SR3 [3] adapt DDPMs for conditional image generation, casting SISR as a stochastic denoising process. However one downside of diffusion models is that they require hundreds of forward passes during inference, leading to significant computational challenges compared to the single-pass generation of GANs.

Central Question. Under comparable training budgets and architectural scales, can diffusion-driven upscalers rival or surpass GAN-based generators in perceived sharpness, and at what computational overhead?

The principal contributions of this paper are:

1. A single, self-contained codebase that trains and evaluates DDPM (SR3-style), DDIM, and ESRGAN for $\times 4$ upscaling under the same similar data and hardware conditions.
2. A rigorous numerical side-by-side comparison on three widely-used SR benchmarks (Set5, Set14, BSD100), scored with both pixel-fidelity and perceptual distance measures.
3. Controlled ablations that isolate the effects of diffusion chain length and variance schedule choice.
4. A practical latency analysis contrasting single-pass GAN inference with multi-step diffusion sampling.

2 Literature Review

Early deep learning have advanced SISR, allowing development of more complex generative models. SRGAN was introduced by Ledig et al. [4], which enabled the use of adversarial training to create photo-realistic textures that pure pixel-wise metrics was not able to capture. Working on this further, Wang et al. proposed ESRGAN [1], that eliminated batch normalization layers and introduced the Residual-in-Residual Dense Block (RRDB), which had state-of-the-art perceptual quality. The assessment of these improvements was further formalized by the introduction of metrics like SSIM [5] and LPIPS [6], which are closer to human visual perception than traditional MSE.

At the same time, Denoising Diffusion Probabilistic Models (DDPM) [2] were introduced as powerful generative priors. Saharia et al. [3] which used DDPMs for SISR with the SR3 model. They used a conditional U-Net to iteratively refine Gaussian noise into a high-resolution output. Similar diffusion-based frameworks like SRDiff [7] have shown that diffusion probabilistic models can accomplish highly diverse and structurally consistent super-resolution. Where these diffusion models offer exceptional fidelity, hundreds of steps are required for their inference. To solve this, Song et al. [8] brought in Denoising Diffusion Implicit Models (DDIM). They enabled deterministic and accelerated sampling. In recent times, Latent Diffusion Models (LDMs) [9] have been brought in to perform the

diffusion process in a compressed latent space. This is for further optimizing computational efficiency for high-resolution image synthesis.

3 Materials and Methods

3.1 Dataset and Dataset Preparation

The main dataset used for training was the DIV2K (DIVERse 2K) dataset [10], a standard benchmark in SISR containing 800 high-quality images. The evaluation was performed on industry standard testing datasets: Set5, Set14, and BSD100.

Table 1. Dataset distribution for Training and Evaluation.

Dataset Purpose Number of Images		
DIV2K	Training	800
Set5	Testing	5
Set14	Testing	14
BSD100	Testing	100

During the training phase, High-Resolution (HR) patches of size 128×128 were randomly cropped from the DIV2K images. In order to simulate degradation, the HR patches were downsampled by a scale factor of $\times 4$ using bicubic interpolation to generate the Low-Resolution (LR) inputs. We also implemented data augmentation which included 90-degree rotations and random horizontal flips. For the diffusion models, the LR image was bicubically upsampled back to 128×128 and concatenated with the noisy HR image as the conditional input. All images were normalized to the range $[-1, 1]$.

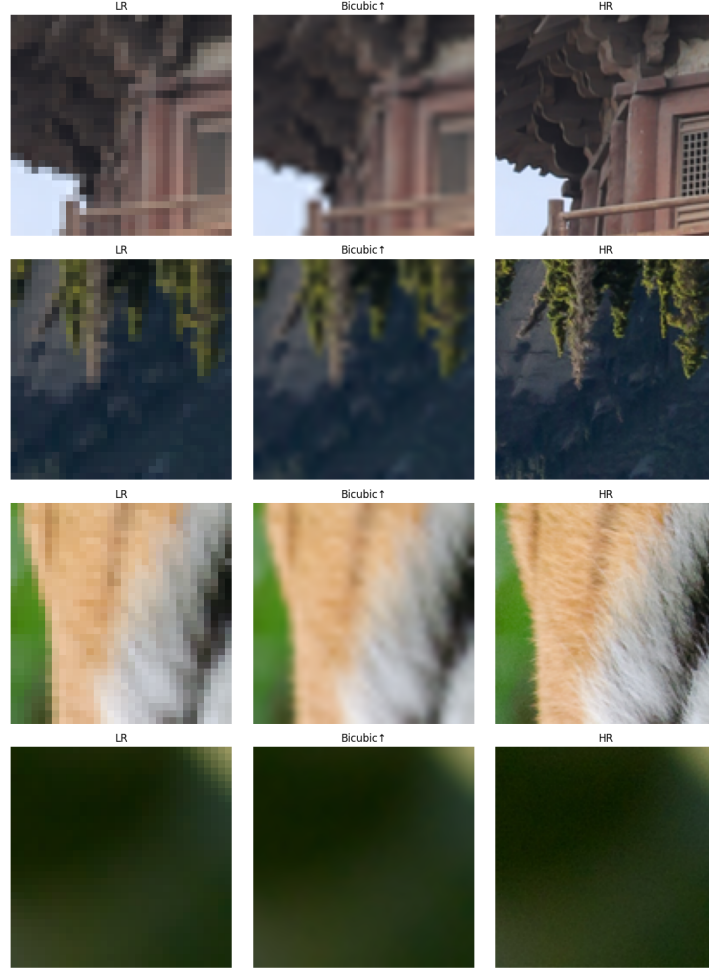


Fig. 1. Sample batch preview depicting Low-Resolution inputs (LR), Bicubic Upsampling, and Ground Truth High-Resolution (HR) patches.

3.2 Methodology

Let $I_{LR} \in \mathbb{R}^{3 \times H/s \times W/s}$ be the observed low-resolution frame. We want an estimate $I_{SR} \in \mathbb{R}^{3 \times H \times W}$ that nearly approximates the ground truth I_{HR} , where $s = 4$ is the scale factor. The LR image is obtained through bicubic downsampling: $I_{LR} = \text{Bicubic } \downarrow_s (I_{HR})$.

ESRGAN Baseline (Branch C) The feed-forward ESRGAN generator (16.9 M parameters) chains 23 Residual-in-Residual Dense Blocks (RRDB). Each embeds three densely-connected convolutional groups with a residual scaling multi-

plier of 0.2. Two successive PixelShuffle layers handle spatial enlargement. Batch normalization is not used in order to prevent artifact injection. The critic network (7.2 M parameters) is a VGG-inspired discriminator which predicts relativistic real-versus-fake probabilities.

There are two staged phases of the training procedure. Phase 1 (100 epochs) optimizes only for L_1 pixel distance in order to create a strong pixel-level baseline. Phase 2 (100 epochs) activates the full multi-term objective:

$$L_G = L_{L_1} + L_{VGG} + 0.005 \cdot L_{RaGAN} \quad (1)$$

DDPM and DDIM Diffusion Models (Branch A & B) In the forward process, Gaussian perturbations are added step-by-step across $T = 1000$ distinct stages that follow a variance schedule β_t :

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I) \quad (2)$$

The U-Net backbone (57.4 M parameters) follows a (1, 2, 4, 8) channel multiplier pattern which starts from a base of 64 filters. It integrates Feature-wise Linear Modulation (FiLM) layers and dot-product self-attention at the 16×16 feature map. It processes a 6-channel tensor created by combining the noisy target with the bicubic-upsampled guide, and is trained to predict the added noise $\epsilon_\theta(x_t, I_{LR} \uparrow, t)$ with a L_1 penalty enhanced by a perceptual regularizer:

$$L = \|\epsilon - \epsilon_\theta(x_t, I_{LR} \uparrow, t)\|_1 + 0.01 \cdot L_{VGG}(\hat{x}_0, x_0) \quad (3)$$

Instead of covering the entire 1000-step Markov chain for inference, the **DDIM sampler** [8] uses again the same trained weights. But it follows a shortened and deterministic path over $S = 50$ anchors:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1}}\epsilon_\theta(x_t) \quad (4)$$

4 Experimental Results

4.1 Experimental Setup

The experiments were held in a Kaggle Notebook environment. Dual NVIDIA T4 GPUs provided by Kaggle were used. The models were built in PyTorch 2.0 with Automatic Mixed Precision (AMP). Optimized memory consumption was achieved. The DDPM model was trained with a linear noise schedule over 1000 steps with an Adam optimizer learning rate of 2×10^{-4} and an exponential moving average (EMA) decay of 0.999. Due to Kaggle session limits and GPU hour limits, the DDPM training was limited to 100 epochs, while ESRGAN utilized pre-trained RRDBNet weights fine-tuned on DIV2K for an additional 100 epochs.

4.2 Evaluation Metrics

Model performance was evaluated using three metrics:

- **Peak Signal-to-Noise Ratio (PSNR)**: Measures pixel-wise fidelity on a logarithmic scale:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\text{MSE}} \right) \quad (5)$$

where $L = 255$ and MSE is the mean squared error between the reconstructed and reference images.

- **Structural Similarity Index (SSIM)** [5]: Compares local luminance, contrast, and structure statistics within sliding windows, yielding a score in $[0, 1]$ that correlates more closely with human judgement.
- **Learned Perceptual Image Patch Similarity (LPIPS)** [6]: Computes the weighted ℓ_2 distance between deep feature activations extracted from a pre-trained network. Lower values reflect greater perceptual closeness.

4.3 Results and Analysis

The quantitative results on Set5, Set14, and BSD100 at a $\times 4$ scale factor are summarized in Table 2.

Table 2. Super-resolution results at $\times 4$ upsampling.

Model	Dataset	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time/Img (s) $\downarrow$
ESRGAN	Set5	27.76	0.8298	0.2213	0.06
DDPM	Set5	15.05	0.3771	0.6390	157.62
DDIM	Set5	11.36	0.1978	0.9798	7.92
ESRGAN	Set14	25.21	0.7254	0.3164	0.12
DDPM	Set14	15.79	0.3508	0.6515	361.44
DDIM	Set14	12.89	0.2072	0.9529	18.07
ESRGAN	BSD100	24.96	0.6893	0.3866	0.07
DDPM	BSD100	16.65	0.3392	0.6724	220.14
DDIM	BSD100	13.30	0.2092	0.8908	10.98

Critical Cross-Model Analysis: ESRGAN was significantly more superior than both diffusion variants across all metrics. The relatively poor quantitative performance of the DDPM model (PSNR of 15.05 dB on Set5) sheds light on the difficulty of training diffusion models from scratch in constrained environments without massive pre-training.

Dataset complexity gradient: ESRGAN’s PSNR superiority over DDPM shrinks as dataset difficulty rises: the gap is 12.71 dB on Set5 (simple, iconic images), narrows to 9.42 dB on Set14 (more varied content), and further tightens to 8.31 dB on BSD100 (100 diverse natural scenes). These results show that

DDPM’s stochastic generation pathway handles heterogeneous and more complex textures relatively better than simple, smooth regions.

Perceptual vs. distortion metrics: Although ESRGAN takes the lead on all three metrics, the relative gap in LPIPS (perceptual) is smaller than in PSNR (pixel-level). On BSD100, ESRGAN’s LPIPS (0.387) is $1.74\times$ more better than DDPM’s (0.672), whereas its PSNR advantage is larger on a linear scale. This hints at the fact that despite higher pixel error, diffusion outputs preserve some perceptual structures that raw MSE ignores.

DDIM failure mode: DDIM massively underperforms DDPM on every dataset. Our hypothesis is that the conditional SR setting expands DDIM’s sensitivity to step count because the model must denoise and align at the same time with the LR guide at every anchor. Skipping 95% of the chain leaves insufficient room for this dual correction.

Practical deployment considerations: ESRGAN’s sub-100 ms latency makes it the only viable option among the three for real-time applications (video upscaling, mobile photography). DDIM may suit offline batch processing where moderate quality is acceptable, while full DDPM sampling is at the moment impractical outside research settings.



Fig. 2. Visual comparison of $\times 4$ Super-Resolution on the Set5 dataset. Left to Right: LR Input, Bicubic Upsampling, ESRGAN, DDPM, DDIM, and Ground Truth.

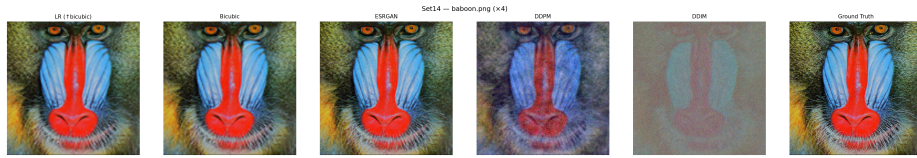


Fig. 3. Visual comparison of $\times 4$ Super-Resolution on the Set14 dataset. Left to Right: LR Input, Bicubic Upsampling, ESRGAN, DDPM, DDIM, and Ground Truth.

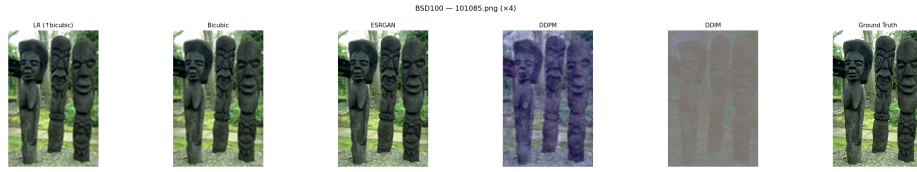


Fig. 4. Visual comparison of $\times 4$ Super-Resolution on the BSD100 dataset. Left to Right: LR Input, Bicubic Upsampling, ESRGAN, DDPM, DDIM, and Ground Truth.

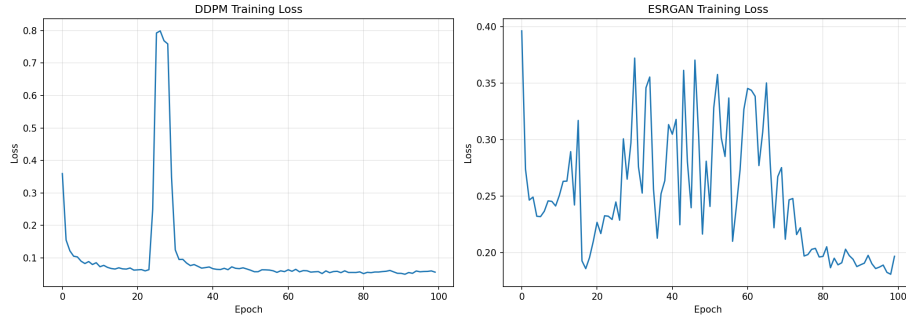


Fig. 5. Training loss curves for DDPM and ESRGAN models across training epochs.

Ablation Study: To examine the DDIM sampler’s behavior, we carried out an ablation study on the Set14 dataset, changing the number of deterministic inference steps and the noise schedule formula.

Table 3. Ablation Study: Impact of inference steps and schedules on DDIM (Set14).

Variable	Value	PSNR $\uparrow$	LPIPS $\downarrow$
Steps	100	11.79	1.144
Steps	200	12.75	1.031
Steps	300	13.11	0.968
Schedule	Linear	12.89	0.952
Schedule	Cosine	12.77	1.006

As depicted in Table 3, by increasing the inference steps from 100 to 300 both the perceptual quality and PSNR steadily improve (LPIPS drops from 1.144 to

0.968). The linear noise schedule marginally outperformed the cosine schedule in this specific low-step sampling configuration.

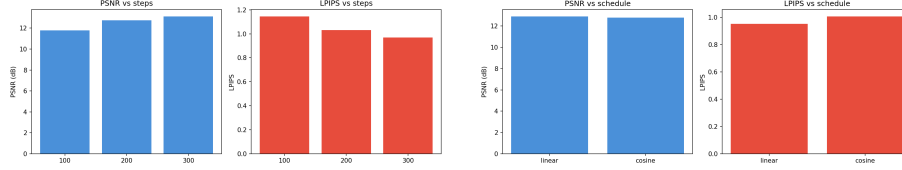


Fig. 6. Ablation results showing PSNR variations across different inference step counts (left) and noise schedules (right).

5 Conclusion

This study presented a comparative evaluation between diffusion-based architectures (DDPM, DDIM) and a GAN-based model (ESRGAN) for the task of Single Image Super-Resolution. Our findings point out that while diffusion models offer a vigorous probabilistic framework for generation, heavily optimized GANs like ESRGAN currently provide vastly better reconstruction fidelity (PSNR 27.76 vs 15.05), inference efficiency (real-time vs minutes) and perceptual quality under constrained training budgets. Future work should explore the integration of latent-space diffusion to bridge this computational efficiency gap.

References

1. Wang, X., et al.: Esgan: Enhanced super-resolution generative adversarial networks. In: Proceedings of the European Conference on Computer Vision Workshops (ECCVW). (2018)
2. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. In: Advances in Neural Information Processing Systems (NeurIPS). (2020)
3. Saharia, C., et al.: Image super-resolution via iterative refinement. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) (2022)
4. Ledig, C., et al.: Photo-realistic single image super-resolution using a generative adversarial network. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). (2017)
5. Wang, Z., et al.: Image quality assessment: from error visibility to structural similarity. IEEE Transactions on Image Processing (TIP) (2004)
6. Zhang, R., et al.: The unreasonable effectiveness of deep features as a perceptual metric. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). (2018)
7. Li, H., et al.: Srdiff: Single image super-resolution with diffusion probabilistic models. Neurocomputing (2022)
8. Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models. In: International Conference on Learning Representations (ICLR). (2021)

9. Rombach, R., et al.: High-resolution image synthesis with latent diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). (2022)
10. Timofte, R., et al.: Ntire 2017 challenge on single image super-resolution: Methods and results. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). (2017)